

Young Couple

Problem 1

- Find the least overall time-consuming work plan for the young couple

Sets

- Persons : $p \in \{Eve, Steven\}$
- Jobs: $j \in \{cleaning, cooking, dishwashing, laundry\}$

Parameters

- Working time for person p to do job j : $Time_{p,j}$

Decision variables

- $x_{p,j}$: 1 if person p performs job j and 0 otherwise.

Model

Objective:

- Objective is working time to be minimized:

$$\sum_p \sum_j Time_{p,j} \cdot x_{p,j}$$

Constraints:

- All jobs must be done exactly once:

$$\sum_p x_{p,j} = 1, \forall j$$

The full model in Julia/JuMP, available with the name

YoungCouple1.jl

from the book web-site, is given below:

```
#####
# Young Couple Assignment, "Mathematical Programming Modelling" (42112)

#####
# Intro definitions
using JuMP
using HiGHS
#####

#####
# PARAMETERS
Jobs = ["clean" "cook" "dishwashing" "laundry"]
J=length(Jobs)
Persons = ["Eve", "Steven"]
P=length(Persons)

Time = [ # Time[p,t]
         4.5      7.8      3.6      2.9;
         4.9      7.2      4.3      3.1
        ]

#####

#####
# Model
YoungCouple = Model(HiGHS.Optimizer)

@variable(YoungCouple, x[1:P, 1:J], Bin) # 1 if person p performs job j

# Minimize work-time
@objective(YoungCouple, Min,
           sum(Time[p,j]*x[p,j] for p=1:P, j=1:J))

# each job should be performed
@constraint(YoungCouple, [j=1:J],
            sum(x[p,j] for p=1:P) == 1)

print(YoungCouple)
#####
```

```

#####
# solve
optimize!(YoungCouple)
println("Termination status: $(termination_status(YoungCouple))")
#####

#####
# Report results
println("-----");
if termination_status(YoungCouple) == MOI.OPTIMAL
    println("RESULTS:")
    println("objective = $(objective_value(YoungCouple))")
    for p in 1:P
        println("$(Persons[p])")
        for j in 1:J
            println("  $(value(x[p,j]))")
        end
    end
else
    println("  No solution")
end
println("-----");
#####

#####
println("Successfull end of $(PROGRAM_FILE)")
#####

```

Comments to the Julia/JuMP program:

- Notice: The binary variables are decision variables which correspond to the decision of who should do the job.

Problem 2

- Find the least time-consuming work plan for the young couple but ensure fairness

Is this a fair solution? No, because Eve is set to perform 3 jobs and Steven only 1. Hence we add another constraint:

Constraints:

- Ensure that Eve and Steven both do the same number of jobs.

$$\sum_j x'_{Eve',j} = \sum_j x'_{Steven',j}$$

Notice that the fair solution requires a higher overall work time than the unfair solution.

The full model in Julia/JuMP, available with the name

`YoungCouple2.jl`

from the book web-site, is given below:

```
#####
# YoungCouple2 Assignment, "Mathematical Programming Modelling" (42112)
#####

#####
# Intro definitions
using JuMP
using HiGHS
#####

#####
# PARAMETERS
Jobs = ["clean" "cook" "dishwashing" "laundry"]
J=length(Jobs)
Persons = ["Eve", "Steven"]
P=length(Persons)

Time = [ # Time[p,t]
         4.5      7.8      3.6      2.9;
         4.9      7.2      4.3      3.1
        ]

#####

#####
# Model
YoungCouple2 = Model(HiGHS.Optimizer)

@variable(YoungCouple2, x[1:P, 1:J], Bin) # 1 if person p performs job j

# Minimize work-time
@objective(YoungCouple2, Min,
           sum(Time[p,j]*x[p,j] for p=1:P, j=1:J))
```

```

# each job should be performed
@constraint(YoungCouple2, [j=1:J],
            sum(x[p,j] for p=1:P) == 1)

# equal job distribution
@constraint(YoungCouple2,
            sum(x[1,j] for j=1:J) == sum(x[2,j] for j=1:J) )

print(YoungCouple2)
#####

#####
# solve
optimize!(YoungCouple2)
println("Termination status: $(termination_status(YoungCouple2))")
#####

#####
# Report results
println("-----");
if termination_status(YoungCouple2) == MOI.OPTIMAL
    println("RESULTS:")
    println("objective = $(objective_value(YoungCouple2))")
    for p in 1:P
        println("$(Persons[p])")
        for j in 1:J
            println("  $(value(x[p,j]))")
        end
    end
else
    println("  No solution")
end
println("-----");
#####

#####
println("Successful end of $(PROGRAM_FILE)")
#####

```